Problem-Solution Fit

Date	01 October 2026
Team ID	SWTID-2026-6912
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to show how FitBuddy addresses the needs, problems, behaviors, limitations, available solutions, and root causes of its target users.

1. CUSTOMER SEGMENT(S) [CS]

• Individuals seeking personalized fitness plans
• Beginners and regular fitness users
• Coaches/admins managing multiple users

9. PROBLEM ROOT / CAUSE [RC]

Lack of an easy, adaptive system that combines user inputs, AI-generated routines, feedback-based updates, and goal-aligned tips.

5. AVAILABLE SOLUTIONS PROS & CONS [AS]

Existing options: generic workout apps, online plans, manual coaching
Pros: accessible, many routines
Cons: may not adapt quickly to feedback or individual goals

2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

• Generic routines may not match goals
• Planning a weekly routine takes effort
• Users may need changes as preferences evolve

7. BEHAVIOR + ITS INTENSITY [BE]

• Users enter age, weight, goal and intensity
• Review a 7-day plan
• Submit feedback to refine the plan
• Use nutrition/recovery tips

3. TRIGGERS TO ACT [TR]

• Starting a fitness goal
• Wanting a structured weekly routine
• Need for a change in workout focus
• Desire for quick nutrition/recovery guidance

8. CHANNELS OF BEHAVIOR [CH]

ONLINE
Web interface, FastAPI API endpoints, interactive HTML/Jinja2 pages

OFFLINE
User's personal workout execution and feedback based on their routine

4. EMOTIONS BEFORE / AFTER [EM]

Before: uncertain, unstructured, time-consuming planning

After: clearer routine, personalized guidance, easier plan refinement

FitBuddy - Team Work Allocation

Team ID: SWTID-2026-6912 **Project:** FitBuddy - AI Fitness Plan Generator using Gemini Models

S.No	Team Member	Assigned Work	Key Deliverables
1	**Pavithra M**	**AI Model Selection & Architecture** Research Gemini models; define the overall FastAPI + Gemini + SQLite architecture; help set up the development environment.	Model selection notes, architecture design, environment setup.
2	**Manisha S**	**Core AI Functionality** Develop workout-plan generation with Gemini Pro, nutrition/recovery tip generation with Gemini Flash, and feedback-based plan updating.	Gemini integration modules, prompts, workout/nutrition generation, plan update logic.
3	**Monica G**	**Backend & Database** Implement FastAPI backend processing, Pydantic input validation, routes, and SQLite/SQLAlchemy data storage.	routes.py, database functions, input processing, user/plan storage.
4	**Monisha Anandan**	**Frontend & Jinja2 Templates** Design the user interface and develop index.html, result.html and all_users.html with responsive CSS and dynamic Jinja2 rendering.	HTML/CSS UI, Jinja2 templates, result display, feedback form, admin view.
5	**Mridula M**	**Testing, Deployment & Documentation** Prepare local deployment, verify endpoints and user flows, test generated plans/feedback updates, and maintain project documentation.	Local deployment, test cases/results, verification, screenshots and documentation.

Work allocation basis:

The responsibilities are assigned in the same order as the names provided and are aligned with the project workflow documented for FitBuddy: model selection and architecture, core functionality, FastAPI/routes and storage, frontend development, and deployment/testing.

Source basis: FitBuddy project description, workflow milestones, technical architecture, core functionality, routes, frontend, and deployment sections.